KidneyShield

Pentoxifylline as a candidate adjunct for chronic kidney disease

Evidence package overview

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Drug and generic forms. Pentoxifylline is a methylxanthine derivative and hemorheologic agent. The cited U.S. DailyMed listing describes a generic 400 mg extended-release oral tablet. Verify current marketed products and the selected product's label before submission [5].

Labeled indication and dosing. The cited extended-release product is labeled for intermittent claudication associated with chronic occlusive arterial disease of the limbs. The usual adult regimen is 400 mg orally three times daily with meals. The label describes reduction to 400 mg twice daily for intolerance and 400 mg once daily for severe renal impairment (creatinine clearance below 30 mL/min). These are instructions for the approved indication, not a validated CKD regimen [5].

Contraindications and warnings. Contraindications include recent cerebral and/or retinal hemorrhage and previous intolerance to pentoxifylline or methylxanthines. Important warnings and precautions concern bleeding, anticoagulant use, theophylline interactions, changes in exposure in renal or hepatic impairment, and interactions with strong CYP1A2 inhibitors. Verify the exact wording against the current product label [5].

Proposed new indication. Evaluate pentoxifylline as an adjunct to standard care in adults with CKD, initially focusing on diabetic CKD with persistent albuminuria or proteinuria. This is a research hypothesis, not an approved indication or a recommendation for routine prescribing.

Difference from the labeled population. The labeled population has intermittent claudication caused by peripheral arterial disease. The proposed population has CKD and may have reduced kidney function, diabetes, cardiovascular comorbidity, polypharmacy, and increased exposure to active metabolites. Efficacy, dosing, and benefit-risk therefore require assessment in the proposed population.

Proposed dosage form and regimen. The candidate formulation is an existing oral extended-release tablet. A CKD-specific dose, frequency, titration schedule, and duration have not been established. Any research regimen must be justified by clinical evidence, pharmacokinetics, renal function, and safety data.

Evidence and principal uncertainty. The PREDIAN randomized trial reported a smaller eGFR decline and reduced urinary albumin excretion with pentoxifylline added to renin–angiotensin system blockade in adults with type 2 diabetes and CKD stages 3–4. Reviews report favorable signals for selected outcomes, but definitive benefit on kidney failure, cardiovascular events, or mortality has not been established [1, 2, 4].

Project contribution. KidneyShield organizes the clinical evidence, labeling, interactions, and research gaps. Published evidence is marked [P], completed and reproducible participant-generated analyses [*], and AI-assisted drafting or organization [AI]. No original statistical analysis is claimed unless separately completed and documented.

A. Rationale

Pentoxifylline is a methylxanthine derivative with hemorheologic effects. The U.S. label describes effects on blood viscosity, erythrocyte flexibility, and experimental leukocyte/neutrophil functions. Its approved indication is intermittent claudication associated with chronic occlusive arterial disease of the limbs, not CKD [5].

KidneyShield evaluates whether the existing oral extended-release formulation merits further investigation as an adjunct to contemporary CKD care. The initial research population is adults with diabetic CKD and persistent albuminuria or proteinuria despite indicated standard treatment. This focus is consistent with PREDIAN but does not establish efficacy across CKD etiologies.

CKD progression depends on etiology, eGFR, albuminuria, blood pressure, metabolic factors, comorbidities, and treatment. Contemporary management includes treatment of the underlying cause and indicated evidence-based interventions such as renin–angiotensin system (RAS) inhibition and SGLT2 inhibitors. A proposed adjunct must demonstrate incremental benefit over contemporary care [6].

Pentoxifylline has been studied because of proposed anti-inflammatory and hemorheologic effects and possible effects on proteinuria. PREDIAN reported a smaller eGFR decline in participants with type 2 diabetes and stage 3–4 CKD receiving pentoxifylline alongside RAS inhibition [1]. A 2016 review included 26 studies and 1,518 participants, but found no evidence in the included trials for hard renal outcomes such as doubling of serum creatinine or chronic dialysis [2]. A 2026 meta-analysis reported favorable pooled estimates for eGFR and urinary albumin excretion; these findings require appraisal of study quality, heterogeneity, and clinical relevance [4].

Intermediate outcomes must not be conflated with kidney failure, transplantation, cardiovascular events, or mortality. A favorable change in eGFR or albuminuria does not itself prove improvement in these patient-important outcomes.

The U.S. Renal Data System reported that care for adults with CKD covered by Medicare fee-for-service cost $141.1 billion in 2023, equivalent to 36% of total costs in that defined coverage group [7]. This establishes healthcare burden, not savings attributable to pentoxifylline. Economic value would require evidence on clinical outcomes, acquisition and monitoring costs, adverse events, and resource use.

Development question: In a defined CKD population, particularly diabetic CKD with persistent albuminuria or proteinuria, does adjunctive pentoxifylline improve kidney outcomes when added to contemporary standard care, with an acceptable safety profile?

B. Risk/Benefit

Potential benefits. Evidence suggests possible effects on eGFR decline, albuminuria/proteinuria, and inflammatory biomarkers. These are signals requiring confirmation, not established benefits for the proposed indication. Generic availability may support access but does not establish effectiveness or cost-effectiveness.

Potential risks. The label describes gastrointestinal and central nervous system adverse effects, bleeding reports with or without NSAIDs, anticoagulants, or platelet aggregation inhibitors, and interactions with theophylline and strong CYP1A2 inhibitors. Occasional angina, hypotension, and arrhythmia are also described [5]. CKD patients may have polypharmacy and altered exposure to active metabolite V, particularly with severe renal impairment. The clinical consequences of increased metabolite exposure are not fully established [5].

Benefit-risk conclusion. Current evidence supports further evaluation, not routine use to slow CKD progression. A compelling case requires replication, clinically meaningful kidney endpoints,

safety characterization in reduced eGFR, comparison with contemporary standard care, and measurement of adherence and discontinuation.

Before a definitive efficacy trial, investigators should complete a source-level evidence synthesis and resolve uncertainties in dose selection and pharmacokinetics. A confirmatory trial should prespecify population, background therapy, endpoints, sample-size assumptions, statistical analysis, safety monitoring, and stopping rules. A short-term biomarker study alone would not establish a reduction in kidney failure or mortality.

C. Non-clinical evidence

The label describes reduced blood viscosity, improved erythrocyte flexibility, and experimental effects on leukocyte deformability and neutrophil adhesion/activation [5]. These actions provide a possible rationale for investigation in microvascular or inflammatory disease but do not establish the mechanism of any clinical kidney effect.

Experimental work has examined pentoxifylline in kidney injury models, including a remnant-kidney study [9]. Before relying on that study, verify the original report's model, species, sample size, dose, route, duration, comparator, kidney outcomes, and adverse findings. Experimental results cannot establish an effective human dose or patient-important benefit.

For each included non-clinical study, record the citation, model, sample size, intervention, comparator, endpoints, numerical results, adverse findings, and limitations. No unpublished experimental result is claimed in this draft. Any participant-generated result must be marked [*] only after completion and documented with methods and data provenance.

D. Safety

Label and contraindications. The cited DailyMed extended-release label is listed as updated in May 2026, with a product-label revision in January 2026. Confirm that it remains the appropriate current label and record the final access date [5]. Contraindications include recent cerebral and/or retinal hemorrhage and previous intolerance to pentoxifylline or methylxanthines.

Bleeding and interactions. The label reports bleeding with or without NSAIDs, anticoagulants, or platelet aggregation inhibitors and increased prothrombin time with vitamin K antagonists. It also describes possible increased theophylline levels, increased exposure with strong CYP1A2 inhibitors such as ciprofloxacin or fluvoxamine, increased concentrations with cimetidine, and possible blood-pressure effects with antihypertensives [5]. These risks warrant medication reconciliation and prospective safety monitoring in a CKD study.

Pharmacokinetics in renal impairment. Pentoxifylline is extensively metabolized, with active metabolites substantially excreted in urine. For a 400 mg three-times-daily regimen, the label reports that active metabolite V AUC and maximum concentration increase approximately 2.4-fold and 2.1-fold in mild-to-moderate renal impairment, and 12.9-fold and 10.6-fold in severe renal impairment. The clinical consequences of increased exposure are unknown [5]. For the labeled indication, the product label reduces dosing to 400 mg once daily when creatinine clearance is below 30 mL/min. This is not a validated CKD-treatment regimen. The label also reports altered exposure in hepatic impairment.

Acute toxicity. The label describes overdose reports involving flushing, hypotension, convulsions, somnolence, loss of consciousness, fever, and agitation. A poison-control-center report involving 44 patients described recovery in all reported patients. These reports do not replace a full formal toxicology review [5].

Carcinogenicity and genotoxicity. Long-term dietary studies in mice and rats used doses up to

450 mg/kg. In female rats, the label reports a statistically significant increase in benign mammary fibroadenomas at the highest dose; relevance to humans is uncertain. The label reports negative results in the described Salmonella Ames, cultured-cell, and mouse micronucleus tests. These assay results do not exclude every possible risk [5].

Reproductive toxicity and chronic exposure. No fetal malformations were observed at the reported doses in rats and rabbits, but increased resorption occurred in rats at the highest reported dose. Adequate, well-controlled studies in pregnant women are lacking [5]. The label describes clinical exposure periods up to 60 weeks for extended-release tablets and 24 weeks for immediate-release capsules; this does not establish safety throughout the longer period potentially needed to assess CKD progression.

Proposed study monitoring. A study should prespecify assessment of eGFR, blood pressure, cardiovascular symptoms, bleeding, gastrointestinal intolerance, relevant blood counts, concomitant medicines, adherence, dose interruptions, and discontinuation. Qualified investigators must define the actual monitoring and stopping procedures. This section summarizes evidence and research considerations, not individualized medical advice.

E. Clinical and/or real-world data/evidence

E.1. Evidence-review approach

This document is an evidence synthesis in progress, not a completed systematic review. A submission-ready review should retain a dated search log, exact search strings, eligibility criteria, screening counts, duplicate handling, extraction rules, risk-of-bias assessments, and records of how conflicting values were resolved.

eGFR change, eGFR slope, urinary albumin excretion, UACR, proteinuria, biomarkers, kidney failure, and mortality are distinct outcomes and should not be treated as interchangeable.

E.2. PREDIAN randomized trial

Navarro-González and colleagues reported PREDIAN, a prospective, randomized, open-label, controlled study of 169 adults with type 2 diabetes and CKD stages 3–4: 82 received pentoxifylline and 87 received control treatment. Both groups received RAS inhibition, and follow-up lasted two years [1].

Eligibility. Reported inclusion criteria were age over 40 years, type 2 diabetes duration of at least eight years, diabetic nephropathy, CKD stages 3–4, urinary albumin excretion above 30 mg/24 hours, stable renal function (serum-creatinine variability below 15% compared with a measurement four to eight weeks earlier), diabetic retinopathy, and ACE-inhibitor or ARB treatment at the maximum recommended dose for more than six months.

Exclusions. Reported exclusions included type 1 diabetes, non-diabetic kidney disease, chronic inflammatory, immunologic, or tumoral disease, recent acute inflammation or infection, institutionalization, immunotherapy or immunosuppression, previous pentoxifylline, specified combinations of RAS-active drugs, severe uncontrolled hypertension, HbA1c above 10%, protocol-defined pregnancy-related conditions, and refusal to participate. Verify the full list against the original article.

Intervention and statistics. The intervention was pentoxifylline 1,200 mg/day (400 mg three times daily) added to RAS inhibition; controls received RAS inhibition alone. The planned sample size of 168 was estimated to provide 80% power to detect a 35% difference in eGFR change at a two-sided 5% significance level, allowing for 5% dropout. Continuous outcomes were summarized using means and standard deviations, except urinary albumin excretion and TNF-α, reported as medians and interquartile ranges. Baseline comparisons used independent-sample t tests, chi-squared

tests, or Fisher exact tests as appropriate. The report describes an intention-to-treat population requiring eligibility, at least one dose, and at least one post-randomization eGFR measurement. Repeated-measures ANCOVA assessed changes in eGFR and urinary albumin excretion, with treatment group and baseline eGFR/albumin excretion as covariates; least-square means, standard errors, and two-sided 95% confidence intervals were estimated. Fisher exact tests and Spearman rank correlations were used for specified analyses [1].

Results. Mean eGFR decline was 2.1 ± 0.4 mL/min/1.73 m^2 with pentoxifylline and 6.5 ± 0.4 with control. The reported between-group difference was 4.3 mL/min/1.73 m^2 (95% CI 3.1–5.5; $P < 0.001$) [1].

The abstract reports percentage change in urinary albumin excretion of +5.7% (95% CI −0.3% to 11.1%) in controls and −14.9% (95% CI −20.4% to −9.4%) with pentoxifylline ($P = 0.001$) [1]. Confirm the exact endpoint definition and numerical values against the full article before submission.

Gastrointestinal symptoms occurred in 18/82 participants (21.9%) in the pentoxifylline group and 9/87 (10.3%) in controls ($P = 0.03$) [1]. The complete adverse-event table, dose reductions, discontinuations, and missing follow-up should be extracted before finalization.

Limitations. PREDIAN was single-center, open-label, and relatively small. It studied a specific diabetic CKD population receiving RAS inhibition and does not establish benefit across CKD etiologies or on top of every component of contemporary care. It was not sufficient to establish reduced kidney failure or mortality.

E.3. 2016 systematic review and meta-analysis

Leporini and colleagues included 26 studies and 1,518 participants in a systematic review and meta-analysis of pentoxifylline in CKD [2]. The authors reported that trials lacked hard renal outcomes such as doubling of serum creatinine or chronic dialysis. They reported reduced proteinuria compared with controls and improvements in selected kidney-function measures in some analyses, but cumulative analyses did not show a clear reduction in urinary albumin excretion. The authors concluded that evidence for improved renal outcomes was inconclusive.

Proteinuria and albumin excretion are related but not identical outcomes. The final package should extract exact pooled estimates, confidence intervals, heterogeneity, included-study characteristics, risk-of-bias assessments, and sensitivity analyses from the original review rather than infer them from secondary summaries.

E.4. Long-term follow-up report

A 2019 report by Muñoz de Morales and colleagues has been cited in the pentoxifylline CKD literature [3]. Verify the original title, design, cohort, follow-up, event counts, covariates, adjustment model, confidence intervals, and limitations before including quantitative claims. Observational or post hoc associations may be affected by residual confounding and selection. No hazard ratio or causal benefit is asserted here.

E.5. 2026 systematic review and meta-analysis

Xu, Xu, and Shen reported a review of 19 studies involving 1,166 participants. The PubMed abstract reports searches of PubMed, Embase, the Cochrane Library, and Web of Science through April 9, 2026; two independent reviewers for selection, extraction, and quality assessment; Stata 15; pooling of continuous outcomes as mean differences (MD) or standardized mean differences (SMD) with 95% confidence intervals; and assessment of heterogeneity using I^2 [4].

Outcome	Abstract estimate	Interpretation
eGFR	7 studies, $N = 547$; MD +4.59 mL/min/1.73 m^2 (95% CI 2.57–6.61)	Favorable pooled estimate; assess heterogeneity and study quality.
UAER	6 studies, $N = 494$; SMD −0.57 (95% CI −1.01 to −0.12)	Suggests lower urinary albumin excretion.
CRP	10 studies, $N = 594$; SMD −0.70 (95% CI −1.08 to −0.31)	Biomarker outcome, not proof of clinical benefit.
TNF-α	5 studies, $N = 246$; SMD −0.68 (95% CI −1.19 to −0.18)	Biomarker outcome; clinical relevance uncertain.
Hemoglobin	9 studies, $N = 424$; SMD +0.51 (95% CI 0.08–0.94)	Verify heterogeneity and clinical relevance.
Serum albumin	7 studies, $N = 355$; MD +0.19 g/dL (95% CI 0.00–0.38)	Verify rounding and full-text results.

These are estimates reported in the abstract, not independently recalculated values. Verify the full publication, endpoint definitions, model choice, heterogeneity, risk of bias, sensitivity analyses, and any corrections. UAER must not be conflated with UACR; no separate UACR result is asserted here.

E.6. Integrated interpretation

The literature supports a research hypothesis, but study populations, CKD etiologies, doses, durations, background therapy, and endpoint definitions vary. PREDIAN does not establish incremental benefit in every patient receiving current multidrug CKD care. Meta-analytic results may be influenced by small studies, heterogeneity, publication bias, or measurement differences. Surrogate outcomes do not prove fewer kidney-failure events or deaths.

The appropriate conclusion is that further evaluation may be warranted if a complete source-level appraisal confirms the findings and supports an acceptable safety profile in the intended population.

E.7. AI disclosure and reproducibility

The final submission should document sources searched, dates, exact search strings, inclusion/exclusion criteria, screening counts, extraction fields, quality-assessment method, and analysis code/data for any original figures or quantitative summaries.

AI tool disclosure should identify the actual tool, model/version, dates of use, tasks, and prompts or parameters required by the challenge. This draft was AI-assisted in organization and writing. Before submission, insert the actual model/version and workflow used, and verify all factual claims against original sources. Do not claim completed systematic-review or statistical methods unless supported by records.

No real-world dataset or original statistical analysis is claimed in this draft. Add participant-generated tables or figures only when the underlying data, code, and methods exist; mark them [*]. No kidney photographs or microscopy images are included.

F. Pharmacogenomic considerations

Guo and colleagues studied CYP1A2 polymorphisms and pentoxifylline/metabolite pharmacokinetics in 46 healthy Chinese volunteers after a single 400 mg dose [8]. This may inform hypotheses about exposure variability, but does not establish genotype-guided dosing for CKD. Verify the variants, genotype groups, endpoints, effect estimates, and limitations in the original paper. Routine genetic testing is not recommended by this evidence package.

For initial research, clinically interpretable stratification variables should take precedence over unvalidated genetic rules: baseline eGFR and CKD stage, albuminuria/proteinuria, diabetes and CKD etiology, background RAS/SGLT2 treatment where indicated, concomitant medicines, age, comorbidities, adherence, and safety parameters.

The label reports bleeding concerns with NSAIDs, anticoagulants, and platelet aggregation inhibitors; increased prothrombin time with vitamin K antagonists; possible theophylline toxicity; possible increased exposure with strong CYP1A2 inhibitors; increased concentrations with cimetidine; and possible blood-pressure effects with antihypertensives [5]. Research should document concomitant medicines and prespecify management of interaction-related risks.

G. Treatment regimen

The proposed use is adjunctive, not a replacement for indicated RAS inhibitors, SGLT2 inhibitors, blood-pressure management, diabetes care, or other evidence-based CKD interventions. The existing evidence includes pentoxifylline added to RAS inhibition; incremental benefit on top of contemporary care remains uncertain.

The candidate formulation is an oral extended-release tablet. For intermittent claudication, the cited label specifies 400 mg three times daily with meals, reduction to twice daily for intolerance, and 400 mg once daily for severe renal impairment (creatinine clearance below 30 mL/min) [5]. These instructions must not be presented as a CKD regimen.

No CKD-specific dose or duration is established. A research regimen must be justified from verified clinical evidence, formulation-specific pharmacokinetics, renal-function-dependent metabolite exposure, and safety. If evidence is insufficient, dose-finding and pharmacokinetic evaluation should precede a definitive efficacy trial.

Potential adherence barriers include multiple daily doses, tablet burden, gastrointestinal symptoms, comorbidities, polypharmacy, and long treatment duration. A less frequent regimen should not be assumed equivalent without evidence. Research should measure adherence, dose interruptions, dose reductions, discontinuation, and patient-reported treatment burden.

H. Additional considerations

Development pathway. First, complete the evidence inventory and risk-of-bias assessment; verify numerical results and identify overlapping reports. Second, assess relevance to contemporary care, including use of SGLT2 inhibitors and other indicated treatments. Third, define the target population, dose, duration, and monitoring strategy from verified clinical and pharmacokinetic evidence. If justified, proceed to prospective randomized evaluation.

Future trial. A randomized, double-blind, placebo-controlled trial could compare a justified pentoxifylline regimen with matching placebo, both added to contemporary standard care. The protocol should define CKD etiology, eGFR and albuminuria ranges, background therapy, primary endpoint, follow-up, sample-size assumptions, statistical analysis, missing-data methods, multiplicity strategy, and safety stopping rules. Secondary outcomes may include albuminuria/proteinuria, patient-reported outcomes, adherence, adverse events, and resource use. A defensible sample size and duration require a specified endpoint, effect size or event rate, variance, and attrition assumptions.

Real-world evidence. Electronic health records, claims, or CKD registries may supplement randomized evidence if access and data quality permit. Reproducible analysis requires the data source and period, eligibility, exposure and comparator definitions, outcomes, confounder adjustment, missing-data methods, and sensitivity analyses. Key risks include confounding by indication, immortal-time bias, treatment misclassification, informative censoring, and differences in access to standard care. No real-world dataset is claimed to have been analyzed here.

Economic and implementation considerations. If clinical benefit is established, assess affordability, monitoring, adverse-event-related utilization, medication burden, and cost-effectiveness. Generic status alone does not establish economic value.

Regulatory and compendia implications. This package does not claim FDA approval of pentoxifylline for CKD. A new indication would require adequate efficacy and safety evidence in the proposed population, a justified formulation and dose, and a complete regulatory-quality evidence package. Compendia consideration would depend on the relevant compendium's evidentiary standards. No use patent is asserted or required.

Participant contribution and provenance. Mark participant-generated work [*] only after it has been completed and documented. For original analyses, provide input-file provenance, code, software environment, analysis decisions, and limitations. Published evidence is [P]; AI-assisted drafting/organization is [AI], not independent scientific evidence.

I. References

References

[1] **[P]** Navarro-González JF, Mora-Fernández C, Muros de Fuentes M, et al. Effect of Pentoxifylline on Renal Function and Urinary Albumin Excretion in Patients with Diabetic Kidney Disease: The PREDIAN Trial. *J Am Soc Nephrol*. 2015;26(1):220–229. doi:10.1681/ASN.2014010012. https://pmc.ncbi.nlm.nih.gov/articles/PMC4279740/.

[2] **[P]** Leporini C, Pisano A, Russo E, D'Arrigo G, de Sarro G, Coppolino G, Bolignano D. Effect of pentoxifylline on renal outcomes in chronic kidney disease patients: A systematic review and meta-analysis. *Pharmacol Res*. 2016;107:315–332. doi:10.1016/j.phrs.2016.03.001. https://pubmed.ncbi.nlm.nih.gov/26995301/.

[3] **[P]** Muñoz de Morales A, et al. 2019 report concerning pentoxifylline and kidney outcomes. *J Nephrol*. 2019;32(4):581–587. doi:10.1007/s40620-019-00607-0. PMID: 30949987. https://pubmed.ncbi.nlm.nih.gov/30949987/. *Verify the exact title, author list, design, and findings before submission.*

[4] **[P]** Xu Y, Xu F, Shen W. The efficacy and safety of pentoxifylline in the treatment of chronic kidney disease: a systematic review and meta-analysis. *Eur J Clin Pharmacol*. 2026;82(6):144. doi:10.1007/s00228-026-04075-x. PMID: 42091733. https://pubmed.ncbi.nlm.nih.gov/42091733/. *Verify full publication details and extracted estimates against the complete article.*

[5] **[P]** U.S. National Library of Medicine. DailyMed: Pentoxifylline extended-release tablet, Rising Pharma Holdings, Inc. Label page updated May 15, 2026; product label revised January 2026. https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=cd840a2c-2a0e-4b2b-84ff-a0406bb6fa37. *Verify the selected product and record access date.*

[6] **[P]** Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. *Kidney Int*. 2024. doi:10.1016/j.kint.2023.10.016. https://kdigo.org/guidelines/ckd-evaluation-and-management/.

[7] **[P]** United States Renal Data System. *2025 USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States*. Chapter 6: Healthcare Expenditures for Persons with CKD. https://usrds-adr.niddk.nih.gov/2025/chronic-kidney-disease/6-healthcare-expenditures-for-persons-with-ckd. *Cost context for a defined population only; does not establish pentoxifylline-related savings.*

[8] **[P]** Guo L, et al. Effects of CYP1A2 genetic polymorphisms on the pharmacokinetics of pentoxifylline and its active metabolites. *BMC Pharmacol Toxicol*. 2026;27:51. doi:10.1186/s40360-026-01106-2. https://link.springer.com/article/10.1186/s40360-026-01106-2. *Verify author list and specific results used.*

[9] **[P]** Lin SL, et al. Experimental remnant-kidney study involving pentoxifylline. *J Am Soc Nephrol*. 2002. doi:10.1097/01.ASN.0000034909.10994.8A. PMID: 12444210. https://pubmed.ncbi.nlm.nih.gov/12444210/. *Verify exact title, author list, model, intervention, and findings before making specific claims.*

Final verification. Verify all source details and numerical claims against original publications. Insert the actual AI model/version, use dates, and prompts or parameters required by the challenge. Mark participant-generated analyses [*] only after completion. This package is a research assessment, not clinical guidance or a claim of FDA approval for CKD.